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	Course Name: Artificial Intelligence Lab	
	Course Code: AI-2101L	Semester: 4 th + 5 th
Lab Final Exam		

1. Download the dataset named car-evaluation-dataset.csv from GCR. Split the dataset into training and testing in 70:30 ratio. The dataset contains the following columns: **(30)**

Column	Description	Type
buying	Buying price of the car	Numerical
maint	Maintenance cost of the car	Numerical
doors	Number of doors	Numerical
persons	Person-carrying capacity	Categorical
lug_boot	Luggage boot size	Categorical
safety	Estimated safety level	Categorical
class	Car evaluation rating (Target)	Categorical

Part 1: Data Preprocessing and Exploration (Using Pandas)

1. **Rename Columns:** Rename the dataset columns to: **2**

["Buying", "Maintenance", "Num_Doors", "Num_Persons",
"Luggage_Boot", "Safety", "Car_Evaluation"]

2. **Dataset Overview:**

- Display data types and non-null value counts. **2**
- Print the number of rows and columns. **1**

3. **Unique Values:** **2** Count and display the number of unique values in each column.

4. **Encode** the all the coloumns which needs encoding e.g Num_Persons coloumn. **3**

5. **Class Distribution Plot:** Create a bar plot to visualize the class distribution of the Car_Evaluation column. Use appropriate labels and titles. **3**

Part 2: Classification Using Scikit-learn Perform the following classification tasks. Encode all categorical variables before training.

1. **K-Nearest Neighbors (KNN):** **3**

- Target: Car_Evaluation

2. **Linear Regression:** **3**

- Target: Car_Evaluation

3. **Logistic Regression ...** **4**

Correct EBM -- > 2

For each model, evaluate and print: 4

- F1-Score
- Accuracy
- Precision
- Recall

Part 3: Performance Comparison Table 1

Create a table summarizing the performance of each model:

Algorithm	Accuracy	Precision	Recall
KNN			
Linear Regression			
Logistic Regression			

Question 2 (20)

You are given the **Mall Customers dataset** containing the following columns:

CustomerID, Gender, Age, Annual Income (k\$), Spending Score (1–100)

Q1: Feature Selection 3

Identify the most suitable columns from the dataset to apply the k-means clustering algorithm. Use only relevant features and briefly justify your selection

Q2: Handling Categorical Data 2

- Check whether the Gender column **2** can be directly used in the k-means algorithm.
- If it cannot be used directly, apply an appropriate preprocessing technique to make it suitable for k-means clustering.

Q3: Data Preprocessing

Perform the following preprocessing steps on the dataset:

- Apply **feature scaling 2** on Annual Income (k\$) and Spending Score (1–100) since they are on different scales.
- Check the dataset for **missing values. 1**

Q4: Model Training

Train a **k-means clustering model 2** on the preprocessed dataset using an appropriate value of **k**.

Q5: Visualize how customers are segmented based on spending behavior using a suitable 2-D scatter plot. **3**

Use:

- Annual Income (k\$) on one axis
- Spending Score (1–100) on the other axis
- Different colors for different clusters

Q6: Age vs Spending Analysis 3

Create a plot that shows how different age groups behave in terms of spending.

After plotting, write 2–3 lines explaining what this graph indicates about customer spending behavior across different age groups.